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SEMICONDUCTOR PRODUCTION APPARATUS SYSTEM

Abstract

A semiconductor production apparatus system includes a substrate-holding hand including a first blade support and a second blade support. The semiconductor production apparatus system includes an image capturer for capturing an image of an inspection target including at least one of substrates, a component of a semiconductor production apparatus and the substrate-holding hand, and a controller for detecting a state of the inspection target based on the captured image captured by the image capturer. The image capturer is arranged in the second blade support.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates to a semiconductor production apparatus system, and in particular to a semiconductor production apparatus system including a substrate-holding hands.

BACKGROUND ART

[0002] Substrate conveyor robots including hands for holding substrates are known in the art. See Japanese Patent Publication No. JP 5303301, for example.

[0003] The above Japanese Patent Publication No. JP 5303301 discloses a robot for conveying substrates in and out. The robot includes a substrate-placing mechanism, a sensing hand, a first arm, and a second arm. The substrate-placing mechanism includes a plurality of hand forks on which substrates are placed. The substrate-placing mechanism is supported by the first arm. Also, the sensing hand includes an optical sensor including a light emitter and a light receiver on its distal end. The sensing hand detects storage states of the substrates. In addition to detecting the storage states, the sensing hand conveys the substrates out. Also, the sensing hand is supported by the second arm.

PRIOR ART

Patent Document

[0004] Patent Document 1: Japanese Patent Publication No. JP 5303301

SUMMARY OF THE INVENTION

[0005] However, since the robot described in the above Japanese Patent Publication No. JP 5303301 detects the storage states of substrates by using the optical sensor including the light emitter and the light receiver, it is considered difficult to detect a state of a part of each substrate that is not included in a detection area of the sensor. Although not stated in the above Japanese Patent Publication No. JP 5303301, in a semiconductor production apparatus for performing at least one of conveyance of a substrate, application of processing to a substrate and storing a substrate, states of components of the semiconductor production apparatus including a deterioration state of a storage of the semiconductor production apparatus or a consumption of a consumable are required to be confirmed in some cases. Since the optical sensor for detecting a placement state of a substrate in the above Japanese Patent Publication No. JP 5303301 does not include light emitters and light receivers that are arranged corresponding to the components of the semiconductor production apparatus, it is considered difficult to detect the states of the components of the semiconductor production apparatus in such a case. Also, although not stated in the above Japanese Patent Publication No. JP 5303301, a state of a substrate-holding hand for conveying a substrate is required to be confirmed in some cases. In such a case, it is difficult for the optical sensor for detecting the placement state of the substrate in the above Japanese Patent Publication No. JP 5303301 to detect a state, such as misalignment, of the substrate-holding hand itself on which the sensor is arranged. For these reasons, it is desired to easily detect a state of an inspection target including at least one of a substrate, a component of a semiconductor production apparatus and a substrate-holding hand.

[0006] The present disclosure is intended to solve the above problems, and one object of the present disclosure is to provide a semiconductor production apparatus system capable of easily detecting a state of an inspection target including at least one of a substrate, a component of a

semiconductor production apparatus and a substrate-holding hand.

[0007] A semiconductor production apparatus system according to one aspect of the present disclosure is a semiconductor production apparatus system for at least one of conveyance of substrates into and out of a semiconductor production apparatus for performing at least one of conveyance of the substrates, application of processing to the substrates and storing the substrates, the semiconductor production apparatus system comprising a substrate-holding hand including a first blade support for supporting a proximal end of a first blade on which one of the substrates is placed, and a second blade support for operating independently of the first blade support and for supporting a proximal end of a second blade on which another of the substrates is placed; an image capturer for capturing an image of an inspection target including at least one of the substrates, a component of the semiconductor production apparatus and the substrate-holding hand; and a controller for detecting a state of the inspection target based on the captured image captured by the image capturer, wherein the image capturer is arranged in the second blade support.

[0008] In the semiconductor production apparatus system according to the aspect of the present disclosure, as discussed above, an image capturer for capturing an image of an inspection target including at least one of the substrates, a component of the semiconductor production apparatus and the substrate-holding hand; and a controller for detecting a state of the inspection target based on the captured image captured by the image capturer are provided. Accordingly, since the inspection target can be captured by the image capturer, it is possible to capture an image of the state of the inspection target in a wider area as compared with a case in which an optical sensor is arranged in a distal end of a hand to detect storage states of substrates. For this reason, in a case in which the inspection target is substrates, states of the inspection target can be easily detected based on the captured image captured by the image capturer. Also, the state of the inspection target can be detected based on the captured image without light emitters and light receivers being arranged corresponding to the inspection target by detecting the state of the inspection target based on the captured image captured by the image capturer. For this reason, in a case in which the inspection target is the component of the semiconductor production apparatus, it is possible to easily detect the state of the inspection target based on the captured image. Also, in a case in which the inspection target is the substrate-holding hand itself, it is possible to easily detect the state of the inspection target based on the captured image. Consequently, it is possible to easily detect the state of the inspection target including at least one of the substrate, the component of the semiconductor production apparatus and the substrate-holding hand.

[0009] According to the present disclosure, it is possible to easily detect a state of an inspection target including at least one of a substrate, a component of a semiconductor production apparatus and a substrate-holding hand.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a block diagram showing an entire configuration of a substrate processing system according to a first embodiment.

[0011] FIG. 2 is a schematic diagram showing configurations of a substrate-conveying robot, a storage enclosure, and a storage of a processor according to the first embodiment.

[0012] FIG. 3 is a perspective view illustrating a first blade support and a second blade support, which are moved together.

[0013] FIG. 4 is a perspective view illustrating the first blade support and the second blade support, which is moved independently of the first blade support.

[0014] FIG. 5 is a front view for illustrating an arrangement of an image capturer in the second blade support.

[0015] FIG. **6** is a side view for illustrating capture of an image of substrates with being stored in a storage.

[0016] FIG. **7** is a view showing an exemplary captured image of the substrates with being stored in the storage.

[0017] FIG. **8** is a side view for illustrating capture of an image of the storage with no substrate being stored.

[0018] FIG. **9** is a view showing an exemplary captured image of the storage with no substrate being stored.

[0019] FIG. **10** is a view illustrating an exemplary indication on a display if an anomaly is detected.

[0020] FIG. **11** is a flowchart illustrating a substrate-conveying method according to the first embodiment.

[0021] FIG. **12** is a flowchart illustrating an inspection-target anomaly detection method according to the first embodiment.

[0022] FIG. **13** is a block diagram showing an entire configuration of a substrate processing system according to a second embodiment.

[0023] FIG. **14** is a schematic view showing a configuration of a substrate-conveying robot according to the second embodiment.

[0024] FIG. **15** is a block diagram showing an entire configuration of a substrate processing system according to a third embodiment.

[0025] FIG. **16** is a front view for illustrating an arrangement of two image capturers.

MODES FOR CARRYING OUT THE INVENTION

[0026] The following description will describe embodiments embodying the present disclosure with reference to the drawings.

First Embodiment

[0027] The following description describes a configuration of a substrate processing system **100** according to a first embodiment with reference to FIGS. **1** to **10**. The substrate processing system **100** is an example of a semiconductor production apparatus system.

[0028] As shown in FIG. **1**, the substrate processing system **100** includes a substrate-conveying robot **101**, a storage enclosure **102**, a plurality of processors **103**, a display **104**, and a control apparatus **105**. The substrate processing system **100** applies processing to a substrate **10**, such as a semiconductor wafer or printed circuit board, for example. The substrate processing system **100** processes a plurality of substrates **10** stored in the storage enclosure **102**. The storage enclosure **102** may also store the substrates **10** to which processing has been applied. The substrate **10** has a roughly disk shape, for example, and is arranged adjacent to each other in a vertical direction and stored in the storage enclosure **102**. The processor **103** applies processing, such as resist coating or etching, to the substrates **10**, for example. The storage enclosure **102** is an example of a semiconductor production apparatus and an example of a storage. The processor **103** is an example of the semiconductor production apparatus.

[0029] The display **104** indicates information on a status of the substrate processing system **100**. Specifically, the display **104** indicates information indicating operating statuses of the plurality of processors **103** and an operating status of the substrate-conveying robot **101**. The display **104** includes, for example, a liquid crystal display. The control apparatus **105** is a higher-level controller that controls the entire substrate processing system **100**. The control apparatus **105** outputs signals for operating the plurality of processors **103** and the substrate-conveying robot **101**. The control apparatus **105** is a computer including a CPU (Central Processing Unit), a RAM (Random Access Memory), a ROM (Read Only Memory), and the like.

[0030] The substrate-conveying robot **101** is a robot for conveying the substrate **10**. The substrate-conveying robot **101** performs at least one of conveying the substrates **10** out of the processors **103** that apply processing to the substrates **10** and conveying the substrates **10** into the processors **103**. The substrate-conveying robot **101** performs at least one of conveying the substrates **10** out of the

storage enclosure **102** for storing the substrates **10** and conveying the substrates **10** into the storage enclosure **102**. The processors **103** includes storages **20** for storing a plurality of substrates **10**. For example, the substrate-conveying robot **101** conveys the substrates **10** that are stored in the storage enclosure **102** to the storage **20** in one of the plurality of processors **103**. Subsequently, the substrate-conveying robot **101** conveys the substrates **10** that have been processed in one of the processors **103** from the storage **20** in the one processor **103** to the storage **20** in another of the plurality of processors **103**. Here, the storage **20** is an example of a component of the semiconductor production apparatus.

[0031] As shown in FIG. 2, the substrate-conveying robot **101** includes a substrate-holding hand **30**, a linear mover **40**, and an upward/downward mover **50**.

[0032] Five blades, which are blades **91**, **92**, **93**, **94** and **95**, are arranged on the substrate-holding hand **30**. One of the substrates **10** are placed on each of the blades **91**, **92**, **93**, **94** and **95**.

Specifically, the blades **91**, **92**, **93**, **94** and **95** are thin plate-shaped supports for supporting the substrates **10**. The blades **91**, **92**, **93**, **94** and **95** have bifurcated distal ends. Also, each of the blades **91**, **92**, **93**, **94** and **95** supports a back surface of an outer edge of the substrate **10**, which has the roughly disk shape, from a Z2-direction side, which is a lower side in the vertical direction. The substrates **10** are placed separately on the blades **91**, **92**, **93**, **94** and **95** one by one. Here, the blades **91**, **92**, **93** and **94** are examples of first blades. Also, the blade **95** is an example of a second blade.

[0033] The substrate-holding hand **30** includes a first blade support **31** and a second blade support **32**. The first blade support **31** supports proximal ends of a plurality of blades **91**, **92**, **93** and **94**. That is, the first blade support **31** supports the proximal end of each of the plurality of blades **91**, **92**, **93** and **94** from a Y1-direction side in FIG. 2. Also, the second blade support **32** supports a proximal end of one blade **95** on which the substrate **10** is placed separately from the plurality of blades **91**, **92**, **93** and **94**. In other words, the second blade support **32** supports the proximal end of the blade **95** from the Y1-direction side in FIG. 2. The second blade support **32** operates independently of the first blade support **31**. In an arrangement of the five blades **91**, **92**, **93**, **94** and **95** in the substrate-holding hand **30**, the blade **91**, the blade **92**, the blade **95**, the blade **93** and the blade **94** are arranged in this order in a Z direction, which is the vertical direction, from an upper side in the vertical direction.

[0034] As shown in FIG. 3, the first blade support **31** in the substrate-holding hand **30** includes a casing **31a** having a substantially rectangular parallelepiped shape. Also, the first blade support **31** includes connection parts **31b** that are connected to the blades **91**, **92**, **93** and **94**. The second blade support **32** includes a casing **32a**, an arm **32b**, and a connection part **32c**. The casing **32a** is arranged on an X1-direction side, which is a lateral side with respect to the first blade support **31**, and has a substantially rectangular parallelepiped shape. The arm **32b** is a flat plate-like part extending from the casing **32a** toward the first blade support **31** in an X2 direction. The arm **32b** extends toward the X2-direction side to be inserted into the casing **31a** of the first blade support **31**. The connection part **32c** is a part in an end part of the arm **32b** in the X2-direction side connected to the blade **95**. That is, the arm **32b** extends toward the X2-direction side so that the connection part **32c**, to which blade **95** is connected, is arranged between connection parts **31b** to which the blades **91** and **92** are connected, and the connection part **31b** to which the blades **93** and **94** of the first blade support **31** are connected. The casing **31a** of the first blade support **31** has a slit-like gap in its central part in the Z direction into which the arm **32b** of the second blade support **32** is inserted. In other words, the casing **31a** of the first blade support **31** has a shape that is obtained by dividing the casing into two parts, which are a part to which the blades **91** and **92** are connected and a part to which the blades **93** and **94** are connected, in the Z direction, which is the vertical direction. Also, Y1-direction sides of the two divided parts of the casing **31a** are connected to each other. Also, the casing **32a** of the second blade support **32** is connected to the linear mover **40**, which will be described later. Here, the casing **31a** in the first blade support **31** is connected to the linear mover **40**. Also, the first blade support **31** changes positions of the blades **91**, **92**, **93** and **94**

in the Z direction. Specifically, the first blade support **31** changes distances between the five blades **91, 92, 93, 94** and **95** from each other in the vertical direction with the blade **95** supported by the second blade support **32** being centered. In other words, the substrate-holding hand **30** changes the distances between the five blades **91, 92, 93, 94** and **95** from each other to agree with arrangement intervals in the vertical direction of the plurality of substrates **10** that are placed in the storage enclosure **102** or in the storage **20** of the processor **103**.

[0035] As shown in FIG. 2, the linear mover **40** includes a first portion **41** and a second portion **42**. The first blade support **31** and the second blade support **32** are connected to the first portion **41**. The linear mover **40** linearly moves the substrate-holding hand **30**. The linear mover **40** linearly moves the second blade support **32** independently of the first blade support **31**. Specifically, the linear mover **40** slidably moves the first blade support **31** and the second blade support **32** in a straight line. The linear mover **40** moves the first blade support **31** and the second blade support **32** together with each other, and also moves the second blade support **32** without moving the first blade support **31**. In other words, the first portion **41** of the linear mover **40** is connected to the casing **31a** of the first blade support **31** and the casing **32a** of the second blade support **32**, and switches between an operation that moves the casing **31a** and the casing **32a** together with each other, and an operation that moves only the casing **32a**.

[0036] Also, the linear mover **40** conveys the substrates **10** in or out by moving the substrate-holding hand **30** to the Y2-direction side from an arrangement in which the first blade support **31** and the second blade support **32** of the substrate-holding hand **30** are arranged on the Y1-direction side of the first portion **41**. The state in which the first blade support **31** and the second blade support **32** are arranged on the Y1-direction side refers to an avoidance state. As shown in FIG. 3, when moving the five blades **91, 92, 93, 94** and **95**, the linear mover **40** moves both the first blade support **31** and the second blade support **32** together with each other. As shown in FIG. 4, when moving only one blade **95**, the linear mover **40** moves only the second blade support **32** to the Y2-direction side from an arrangement in which both the first blade support **31** and the second blade support **32** are arranged on the Y1-direction side. In other words, the substrate-holding hand **30** switches between conveyance of the five substrates **10** together and conveyance of only one substrate **10**. Here, the Y1 and Y2 directions are examples of movement directions of the second blade.

[0037] The first portion **41** rotates about the Z direction as a rotation axis with respect to the second portion **42** in the linear mover **40**. That is, the first portion **41** rotates in an XY plane. When the first portion **41** rotates, the substrate-holding hand **30** correspondingly rotates about the Z direction as the rotation axis. Here, when the first portion **41** is rotated, the first blade support **31** and the second blade support **32** of the substrate-holding hand **30** are positioned in the avoidance state. The linear mover **40** includes, for example, a servomotor as a drive mechanism. A driving force of the servomotor is used as power for moving the substrate-holding hand **30** through a belt-and-pulley transmission, a ball-and-screw transmission, or the like in the linear mover **40**. The linear mover **40** includes an encoder for acquiring a rotational speed of the servomotor. Operations of the linear mover **40** are controlled by the controller **70**.

[0038] As shown in FIG. 2, the upward/downward mover **50** moves the linear mover **40** upward/downward. The upward/downward mover **50** moves the substrate-holding hand **30** upward/downward by moving the linear mover **40** upward/downward. Specifically, the upward/downward mover **50** is arranged to extend in the Z direction, which is the vertical direction. The second portion **42** of the linear mover **40** is connected to the upward/downward mover **50**. Accordingly, the linear mover **40** and the substrate-holding hand **30** are moved together with each other upward/downward by moving second portion **42** upward/downward by using the upward/downward mover **50**. The upward/downward mover **50** includes, for example, a servomotor as a drive mechanism. A driving force of the servomotor is used as power for moving the substrate-holding hand upward/downward through a belt-and-pulley transmission, a ball-and-

screw transmission, or the like in the upward/downward mover **50**. The upward/downward mover **50** includes an encoder for acquiring a rotational speed of the servomotor. Similar to the linear mover **40**, operations of the upward/downward mover **50** are controlled by the controller **70**. [0039] As shown in FIG. 5, in the first embodiment, an image capturer **60** is arranged in the second blade support **32**. The image capturer **60** is arranged on the second blade support **32** with being spaced away from the connection part **32c** between the second blade support **32** and the blade **95**. In other words, the image capturer **60** is arranged on the second blade support **32** with being also spaced away from the first blade support **31**, which supports the plurality of blades **91**, **92**, **93**, **94** and **95**. Specifically, the image capturer **60** is arranged in a part of the second blade support **32** on an outside with respect to an edge of the blade **95** in the X1 direction as viewed in the Y1 and Y2 directions, which are the movement directions of the blade **95** when the substrate **10** is conveyed in or out to be spaced away from the connection part **32c**. The image capturer **60** is spaced away from the connection part **32c** to prevent influences of heat generated in the image capturer **60** on operations of the connection part **32c** and the first blade support **31**. In more detail, the image capturer **60** is arranged above the casing **32a** in the second blade support **32**. The image capturer **60** is arranged above the casing **32a** to capture an image on the Y2-direction side, which is a blade **95** movement direction side. Also, the image capturer **60** is arranged on a Z1-direction side, which is an upper side in the vertical direction, with respect to the blade **95**. That is, the image capturer **60** is positioned above a placement surface on which the substrate **10** is placed in the blade **95**. The image capturer **60** moves together with the second blade support **32**. In other words, the image capturer **60** is moved together with the second blade support **32** by operation of the linear mover **40** and the upward/downward mover **50**.

[0040] The image capturer **60** captures an image of the inspection target including at least one of the substrates **10**, components of the storage enclosure **102** and the processor **103**, and the substrate-holding hand **30** based on control by the controller **70**. The captured image P captured by the image capturer **60** is output to the controller **70**. The image capturer **60** includes a two-dimensional camera such as a CCD (Charge Coupled Device) image sensor or CMOS (Complementary Metal Oxide Semiconductor) image sensor, including a plurality of imaging elements, for example. The image capturer **60** may include a three-dimensional camera.

[0041] The controller **70** controls operations of the substrate-conveying robot **101**. The controller **70** is, for example, a computer including a CPU, RAM, ROM, and the like. The controller **70** includes a storage device that includes a flash memory, such as an SSD (Solid State Drive). The controller **70** controls the operations of the parts of the substrate-conveying robot **101** based on programs and parameters previously stored in the storage device.

[0042] The communicator **80** communicates with the plurality of processors **103**. The communicator **80** also communicates with the higher-level control apparatus **105**. The communicator **80** includes a communication module for communication through LAN (Local Area Network) and the like. In other words, the controller **70** communicates with the outside of the substrate-conveying robot **101** through the communicator **80**.

Detection of State of Inspection Target

[0043] The following description describes detection of a state of the inspection target by using the controller **70** with reference to FIGS. 6 to 9.

[0044] In the first embodiment, the controller **70** detects the state of the inspection target based on the captured image P captured by the image capturer **60**. The inspection target includes at least one of the substrates **10** that are placed in the storage **20** and the storage enclosure **102**, the components of the storage enclosure **102** and the processor **103**, and the substrate-holding hand **30**. For example, the component of the storage enclosure **102** is a part of the storage enclosure **102** on which the substrate **10** is placed. The component of the processor **103** is equipment, such as the storage **20**, and a consumable in the processor **103**. In other words, the controller **70** performs detection of placement states of the substrates **10** in the storage **20** and the storage enclosure **102**,

detection of an anomaly or deterioration of the components in equipment forming the storage enclosure **102** and the processor **103**, detection of an anomaly or deterioration of the substrate-holding hand **30**, and detection of a consumption degree of a consumable in each part in the substrate processing system **100** including the processor **103** by applying image analysis processing to the captured image P captured by the image capturer **60**.

[0045] As shown in FIG. **6**, the image capturer **60** captures an image of the substrates **10** as the inspection target with being stored in the storage **20** included in the processor **103**, for example. In other words, the controller **70** detects the placement states of the substrates **10** in the storage **20** as the states of the inspection target. Specifically, in a case in which a plurality of substrates **10** are conveyed out of the storage **20** of the processor **103**, the controller **70** first captured an entire image of the plurality of substrates **10** with being stored in the storage **20**. In this case, if size of the storage **20** in the Z direction is larger than a field of view of the image capturer **60**, the controller **70** operates the upward/downward mover **50** and captures a plurality of images by using the image capturer **60**. In this case, both the first blade support **31** and the second blade support **32** of the substrate-holding hand **30** are positioned at positions of the avoidance state, which are on the Y2-direction side.

[0046] As shown in FIG. **7**, the controller **70** detects, based on the captured image P of the substrates **10** captured with being stored in the storage **20**, the placement states of the substrates **10** in the storage **20**. For example, the controller **70** detects placement positions of the substrates **10** in the storage **20** by detecting the substrates **10** in the captured image P by using image analysis. The controller **70** applies the image analysis to the captured image P to detect shapes extending in the horizontal plane of the detected substrates **10** and deformed shapes warped with respect to the horizontal plane.

[0047] The controller **70** controls movement of the substrate-holding hand **30** based on the placement states of the substrates **10** detected based on the captured image P. Specifically, based on the detected placement states of the substrates **10**, operations of the linear mover **40** and the upward/downward mover **50** are controlled so as to place the substrates **10** onto the blades **91**, **92**, **93**, **94** and **95**. In other words, the controller **70** controls movement paths of the blades **91**, **92**, **93**, **94** and **95** based on the detected placement states of the substrates **10**. Also, if detecting a deformed substrate **10** that cannot be conveyed out or in based on the detected placement state of the substrate **10**, the controller **70** outputs an anomaly detection signal indicating that an anomaly occurs in conveyance of the substrate in or out to the outside through the communicator **80**. The controller **70** may stop the operations of the substrate-holding hand **30** if the anomaly occurs in conveyance of the substrate in or out. Alternatively, if the anomaly occurs in conveyance of the substrate in or out, the controller **70** may continue the operations for conveyance of the substrate **10** in or out while avoiding the substrate **10** that is detected as anomalous.

[0048] As shown in FIG. **8**, the image capturer **60** captures an image of the component of the processor **103** as the inspection target. The controller **70** captures a predefined number of images of a plurality of inspection targets that are previously set in maintenance that is periodically conducted at a predetermined interval, such as once a day, for example. Also, in the maintenance process, the controller **70** captures images of a plurality of inspection targets by changing a position of the image capturer **60** by operating the linear mover **40** and the upward/downward mover **50**. Even in a case in which an inspection target is not positioned in a conveyance path of the substrate **10**, the controller **70** controls operations of the linear mover **40** and the upward/downward mover **50** to move the second blade support **32** so as to position the image capturer **60** at a position for capturing an image of the inspection target.

[0049] For example, the image capturer **60** captures an image of the storage **20** as the inspection target with no substrate **10** being stored. In the processor **103**, the storage **20** stores a plurality of substrates **10**. The plurality of substrates **10** are arranged adjacent to each other in the Z direction, which is the vertical direction, with being spaced at a predetermined interval away from each other

in the storage **20**. The storage **20** includes a plurality of substrate placement parts **21** on which the substrates **10** are placed. The substrate placement parts **21** protrude from inner lateral surfaces of the storage **20** in the horizontal directions.

[0050] Here, the substrate placement parts **21** deteriorate with repeated use. Specifically, an anomaly such as chipping or cracking occurs in the substrate placement part **21** in some cases. Also, an anomaly of misalignment of the substrate placement part **21** occurs in some cases. In these cases, an anomaly such as misalignment of the substrate **10** placed in the storage **20** occurs.

[0051] To address this, the controller **70** detects the anomaly in the substrate placement part **21** by comparing a reference image, which is previously set, with the captured image P captured by the image capturer, as shown in FIG. **9**. The controller **70** detects an anomalous area in the captured image P by performing pattern matching processing using the reference image as a template, for example. The reference image is captured by the image capturer **60**, and is stored in the storage device by the controller **70** every when a part of the storage **20** is replaced, or when the storage is subjected to maintenance/inspection, for example.

[0052] Here, when capturing an image of the component of the processor **103** as the inspection target, the controller **70** brings the image capturer **60** in proximity to the inspection target. In other words, the second blade support **32** is brought in proximity to the inspection target so that the image capturer **60** is brought in proximity to the inspection target. Also, at this image capture, the controller **70** does not bring the first blade support **31** in proximity to the inspection target. In other words, when capturing the image of the inspection target, the controller **70** brings the second blade support **32** on which the image capturer **60** is arranged in proximity to the inspection target while spacing the first blade support **31** further away from the inspection target than the second blade support **32**. Specifically, when capturing an image of the storage **20** of the processor **103**, the controller **70** brings only the second blade support **32** in proximity to the inspection target while keeping the first blade support **31** at the position of the avoidance state. As a result, while the captured image P in FIG. **8** includes the blades **91**, **92**, **93**, **94** and **95**, the captured image P in FIG. **9** does not include the blades **91**, **92**, **93** and **94** but includes only the blade **95**. Also, when an anomaly is detected in the component of the processor **103** that is the inspection target, the controller **70** outputs an anomaly detection signal indicating that the anomaly is detected to the processor **103** through the communicator **80**.

[0053] As shown in FIG. **7**, the captured image P captured by the image capturer **60** includes the substrate-holding hand **30**. The controller **70** detects a state of the substrate-holding hand **30** based on the captured image P. Specifically, the captured image P includes the blades **91**, **92**, **93**, **94** and **95** of the substrate-holding hand **30**. The controller **70** captures a predefined number of images of the blades **91**, **92**, **93**, **94** and **95** of the substrate-holding hand **30** as the plurality of inspection targets that are previously set in maintenance that is periodically conducted at a predetermined interval, such as once a day, for example. Subsequently, the controller **70** detects the anomaly of misalignment, chipping or the like of any of the blades **91**, **92**, **93**, **94** and **95** based on the captured image P. When an anomaly of the substrate-holding hand **30**, which is the inspection target, is detected, the controller **70** outputs an anomaly detection signal indicating that the anomaly is detected to the processor **103** through the communicator **80**.

[0054] As shown in FIG. **10**, for example, in the substrate processing system **100**, an anomaly detection indication **104a**, which indicates that the anomaly is detected, is indicated on the display **104** of the substrate processing system **100** in response to the anomaly detection signal output from the controller **70**. FIG. **10** is a view illustrating an exemplary indication when an anomaly of the storage **20** is detected in one of the plurality of processors **103**. The substrate processing system **100** may also stop the operations of the substrate-conveying robot **101** and the processor **103** when the anomaly detection signal is output. Also, the controller **70** may store the captured image P in which the anomaly is detected in the storage device. In this case, a plurality of captured images P that include the captured image P in which the anomaly is detected may be stored as video.

Processing for Controlling Substrate-Conveying Method

[0055] The following description will describe operations in a substrate-conveying method by the substrate-conveying robot **101** with reference to FIG. **11**. In this substrate-conveying method, the states of the substrates **10** that are placed in the storage **20** is detected as the inspection target of the substrates **10**. Here, the following description describes operations of conveying the substrates **10** out of the storage **20**. Processing for controlling the substrate-conveying method is executed by the controller **70**.

[0056] In step **S1**, images of the substrates **10** that are stored in the storage **20** are first captured while moving the image capturer **60** by operating the linear mover **40** and the upward/downward mover **50**. In this step, a plurality of captured images **P** are captured by moving the image capturer **60** in the vertical direction by using the upward/downward mover **50** to capture images of all the substrates **10** that are stored in the storage **20**.

[0057] Subsequently, in step **S2**, image analysis is performed on the captured image **P** to detect the states of the substrates **10** that are placed in the storage **20**. Specifically, based on the captured image **P**, the placement position and shape of the substrate **10** in the storage **20** are detected as the placement state of the substrate **10**.

[0058] Subsequently, in step **S3**, it is determined whether a state of any of the substrates **10** detected in the storage **20** includes an anomaly. If it is determined that the state of any of the substrates **10** includes the anomaly, the procedure goes to step **S4**. If it is not determined that the state of any of the substrates **10** includes any anomaly, the procedure goes to step **S5**.

[0059] In step **S4**, an anomaly detection signal indicating that the anomaly is detected is output. Also, in step **S5**, the substrates **10** are conveyed out of the storage **20** by controlling operations of the linear mover **40** and the upward/downward mover **50** based on the states of the substrates **10** detected in step **S2**. The substrates **10** that are conveyed out of the storage **20** may be conveyed into the storage enclosure **102** for storing the substrate **10**, or may be conveyed into the storage **20** of a processor **103** other than a processor **103** that stored the substrates **10**.

Method for Detecting Anomaly in Inspection Target

[0060] The following description describes a method for detecting an anomaly in components of the processor **103** and the substrate-holding hand **30** with reference to FIG. **12**. For example, processing for controlling the method for detecting the anomaly is periodically executed in the maintenance at a predetermined interval, such as once a day. The processing for controlling the method for detecting the anomaly in the inspection target is executed by the controller **70**.

[0061] First, in step **S11**, an image of the inspection target is captured by the image capturer **60**. For example, the inspection target is the storage **20** with no substrate **10** being placed. In a case in which the processor **103** includes a plurality of storages **20**, images of the plurality of storages **20** are captured. Also, an image of the storage **20** in each of the plurality of processors **103** is captured. Also, the inspection target may be the substrate-holding hand **30**. Subsequently, the captured image **P** is acquired by the image capture by using the image capturer **60**.

[0062] Subsequently, in step **S12**, an anomaly of the inspection target is detected based on the captured image **P**. Specifically, the anomaly of the inspection target is detected by comparing a reference image, which is previously set, with the captured image **P**, which is acquired in step **S11**. Here, in a case in which a plurality of inspection targets are included, a plurality of types of reference images are previously set and stored corresponding to the plurality of inspection targets. The reference image is, for example, a captured image **P** that is captured every when the inspection target is subjected to maintenance or is replaced. The reference image may be a captured image **P** captured in previous maintenance.

[0063] Subsequently, in step **S13**, it is determined whether the anomaly is detected in the inspection target. If the anomaly is detected in the inspection target, the procedure goes to step **S14**. If the anomaly is not detected in the inspection target, the processing for controlling the method for detecting the anomaly in the inspection target ends.

[0064] In step S14, based on the detected anomaly, an anomaly detection signal indicating that the anomaly is detected is output. The anomaly detection signal is transmitted to the higher-level control apparatus **105** of the substrate processing system **100** through the communicator **80**, for example. Subsequently, the anomaly detection indication **104a**, which is information indicating that the anomaly is detected, is indicated on the display **104** by the control apparatus **105**.

Advantages of First Embodiment

[0065] In the first embodiment, the following advantages are obtained.

[0066] The substrate processing system **100** includes the image capturer **60** for capturing an image of the inspection target including at least one of the substrates **10**, components of the storage enclosure **102** and the processor **103**, and the substrate-holding hand **30**; and the controller **70** for detecting a state of the inspection target based on the captured image P captured by the image capturer **60**. Accordingly, since the inspection target can be captured by the image capturer **60**, it is possible to capture an image of the state of the inspection target in a wider area as compared with a case in which an optical sensor is arranged in a distal end of a hand to detect storage states of substrates **10**. For this reason, in a case in which the inspection target is substrates **10**, states of the inspection target can be easily detected based on the captured image P captured by the image capturer. Also, the state of the inspection target can be detected based on the captured image P without light emitters and light receivers being arranged corresponding to the inspection target by detecting the state of the inspection target based on the captured image P captured by the image capturer **60**. For this reason, in a case in which the inspection target is the storage enclosure **102** and a component of the processor **103**, it is possible to easily detect the state of the inspection target based on the captured image P. Also, in a case in which the inspection target is the substrate-holding hand **30** itself, it is possible to easily detect the state of the inspection target based on the captured image P. Consequently, it is possible to easily detect the state of the inspection target including at least one of the substrates **10**, components of the storage enclosure **102** and the processor **103**, and the substrate-holding hand **30**.

[0067] The substrate processing system **100** includes the substrate-holding hand **30** including the first blade support **31** for supporting proximal ends of the blades **91**, **92**, **93** and **94** on which the substrates **10** are placed, and the second blade support **32** for operating independently of the first blade support **31** and for supporting a proximal end of the blade **95** on which another of the substrates **10** is placed. Also, the image capturer **60** is arranged in the second blade support **32**. Accordingly, in a case in which the inspection target is the substrate **10**, and at least one of the storage enclosure **102** and the component of the processor **103**, since the image capturer **60** arranged in the second blade support **32** can be moved to be brought in proximity to the inspection target, an enlarged image of the inspection target can be captured in detail by the image capturer **60**. For this reason, it is possible to accurately detect the state of the inspection target based on the captured image P of the inspection target captured in detail. Also, since the second blade support **32** operates independently of the first blade support **31**, it is possible to prevent the first blade support **31**, and the blades **91**, **92**, **93** and **94** supported by the first blade support **31** from being included in the captured image P. Consequently, it is possible to prevent that the first blade support **31**, and the blades **91**, **92**, **93** and **94** included in the captured image P make detection of the state of the inspection target difficult.

[0068] The image capturer **60** moves together with the second blade support **32**. Accordingly, the image capturer **60** can be moved by moving the second blade support **32**. Consequently, it is possible to prevent an apparatus configuration from becoming complicated as compared with case in which a configuration for moving the image capturer **60** is provided separately from a configuration for moving the second blade support **32**.

[0069] The image capturer **60** is arranged on the second blade support **32** with being spaced away from the connection part **32c** between the second blade support **32** and the blade **95**. Here, heat generated in the image capturer **60** causes an anomaly of an operation of the substrate-holding hand

30 in some cases. To address this, in the first embodiment, since the image capturer **60** is arranged on the second blade support **32** with being spaced away from the connection part **32c** between the second blade support **32** and the blade **95**, it is possible to prevent that the heat generated in the image capturer **60** causes the anomaly of the operation of the substrate-holding hand **30**. Also, in a case in which the blades **91**, **92**, **93** and **94** supported by the first blade support **31**, and the blade **95** supported by the second blade support **32** are arranged adjacent to each other in the vertical direction as in the first embodiment, the image capturer **60** can be spaced away from both each connection part **31b** of the first blade support **31** and the connection part **32c** of the second blade support **32** by spacing the image capturer **60** away from the connection part **32c**. Accordingly, it is possible to prevent that the heat generated in the image capturer **60** causes an anomaly both in each connection part **31b** of the first blade support **31** and the connection part **32c** of the second blade support **32**. For example, in a case in which the first blade support **31** is configured, as in the first embodiment, to support the proximal ends of the plurality of blades **91**, **92**, **93** and **94** and to change positions of the plurality of blades **91**, **92**, **93** and **94** in the vertical direction, it is possible to prevent that the heat generated in the image capturer **60** causes an anomaly of the movement of the blades **91**, **92**, **93** and **94** of the first blade support **31**.

[0070] The image capturer **60** is arranged in a lateral part of the second blade support **32** on an outside with respect to the edge of the blade **95** as viewed in the movement direction of the blade **95** when the substrate **10** is conveyed in or out to be spaced away from the connection part **32c**. Accordingly, since the image capturer **60** is arranged on the outside with respect to the edge of the blade **95**, the image capturer **60** can be positioned at a position spaced sufficiently away from the connection part **32c**. Consequently, it is possible to effectively prevent that the heat generated in the image capturer **60** causes the anomaly of the operation of the substrate-holding hand **30**. Also, since the image capturer **60** is arranged in the lateral part on the outside with respect to the edge of the blade **95**, an image of the blade **95** itself, which is supported by the second blade support **32**, can be captured by the image capturer **60**. Consequently, it is possible to detect an anomaly of misalignment or the like of the blade **95** based on the captured image P captured by the image capturer **60**.

[0071] The image capturer **60** captures images of the inspection target including at least the substrates **10** with being stored in the storage **20** included in the processor **103** and the storage enclosure **102**; and the controller **70** detects, based on the captured images P of the substrates **10** being stored in the storage **20** and the storage enclosure **102**, the placement states of the substrates **10** in the storage **20** and the storage enclosure **102**. Here, in a case in which optical sensors are arranged in distal ends of the blades **91**, **92**, **93**, **94** and **95** on which the substrates **10** are placed to confirm placement states of the substrates **10**, signal lines necessarily extend to the distal ends of the blades **91**, **92**, **93**, **94** and **95** to be connected to the sensors. Since the blades **91**, **92**, **93**, **94** and **95** on which the substrates **10** are placed typically have a relatively thin shape, work for installation of the signal lines is a burden to workers. Contrary to this, in the first embodiment, the controller **70** detects, based on the captured images P of the substrates **10** being stored in the storage **20** and the storage enclosure **102**, the placement states of the substrates **10** in the storage **20** and the storage enclosure **102**. Accordingly, the placement states of the substrates **10** can be detected based on the captured images P captured by the image capturer **60** provided to the second blade support **32**, which supports the proximal end of the blade **95**, without optical sensors arranged in the distal ends of the blades **91**, **92**, **93**, **94** and **95**. Consequently, since the signal lines are not necessarily installed to extend to the distal ends of the blades **91**, **92**, **93**, **94** and **95**, and it is possible to reduce the burden to the workers.

[0072] The substrate-holding hand **30** includes a plurality of blades **91**, **92**, **93** and **94**, which are supported by the first blade support **31**, and one blade **95**, which is supported by the second blade support **32**; and the image capturer **60** is arranged in the second blade support **32**, which supports the proximal end of the one blade **95**. Accordingly, since the image capturer **60** is arranged in the

second blade support **32**, which supports the proximal end of one blade **95**, the second blade support **32** can be smaller than the first blade support **31**, which supports the proximal ends of the plurality of blades **91**, **92**, **93** and **94**. For this reason, since the second blade support **32** can be moved to bring the image capturer **60** into a relatively narrower area as compared with a case in which the image capturer **60** is provided to the first blade support **31**, it is possible to bring the image capturer **60** in proximity to the inspection target that is arranged in the relatively narrow area to capture an image of the inspection target. Consequently, since the captured image P of the inspection target that is arranged in the relatively narrow area can be acquired in detail, it is possible to accurately detect a state of the inspection target even in the case in which the target is arranged in the relatively narrow area.

[0073] The substrate processing system **100** includes the linear mover **40** for linearly moving the substrate-holding hand **30**, and the linear mover **40** linearly moves the second blade support **32** independently of the first blade support **31**. Accordingly, since the linear mover **40** can slidably move the substrate-holding hand **30** in a straight line, it is possible to stably move the substrate-holding hand **30** with the substrate-holding hand being supported from the lower side in the vertical direction. Also, in a case in which the substrate-holding hand **30** is linearly moved by a link mechanism including a horizontal or vertical multi-joint robot arm, its links are necessarily folded for linear movement, and as a result physical interference in a transverse direction that intersects with the movement direction is necessarily taken into account. Contrary to this, in the case in which the substrate-holding hand **30** is linearly moved by the linear mover **40**, the physical interference in the transverse direction can be prevented, and as a result it is possible to linearly move the substrate-holding hand **30** to bring the substrate-holding hand into a narrower area as compared with the case of the link mechanism including the robot arm. Consequently, even when the inspection target is located in a relatively narrow position, since the image capturer **60** can be brought in proximity to the inspection target to capture the captured image P in detail, it is possible to accurately detect the state of the inspection target.

[0074] The upward/downward mover **50** for moving the linear mover **40** upward/downward is provided, and the upward/downward mover **50** moves the substrate-holding hand **30** upward/downward by moving the linear mover **40** upward/downward. Accordingly, the substrates **10** can be conveyed between storages **20** having different height positions in the vertical direction by the substrate-holding hand **30**. Also, even in a case in which inspection targets are positioned at positions different from each other in the vertical direction, since the substrate-holding hand **30** can be moved upward/downward by the upward/downward mover **50**, the image capturer **60** can be brought in proximity to the inspection targets at the different positions in the vertical direction to capture images of the inspection targets. Consequently, it is possible to accurately detect the states of the inspection targets even in the case in which the inspection targets are positioned at the different positions in the vertical direction.

Second Embodiment

[0075] The following description describes a configuration of a substrate processing system **200** according to a second embodiment of the present disclosure with reference to FIGS. **13** and **14**. An image capturer **260** is arranged on a second blade support **232** including a plurality of blades **292**, **293**, **294** and **295** in this second embodiment dissimilar to the aforementioned first embodiment in which the image capturer **60** is arranged in the second blade support **32** including one blade **95**. The same components as those of the aforementioned first embodiment are denoted by the same reference numerals, and their description is omitted.

[0076] As shown in FIG. **13**, the substrate processing system **200** according to the second embodiment includes a substrate-conveying robot **201**. Also, the substrate-conveying robot **201** includes a substrate-holding hand **230**, a first arm **241**, a second arm **242**, an upward/downward mover **250**, and an image capturer **260**. The substrate-holding hand **230** includes a first blade support **231** and a second blade support **232**. The upward/downward mover **250** includes an

upward/downward moving shaft **251**. The substrate processing system **200** is an example of a semiconductor production apparatus system.

[0077] As shown in FIG. **14**, the substrate-conveying robot **201** is a horizontal multi-joint robot dissimilar to the substrate-conveying robot **101** according to the first embodiment. In other words, the substrate-conveying robot **201** includes the first arm **241** and the second arm **242**, which are horizontal multi-joint robot arms, instead of the linear mover **40**. The first arm **241** and the second arm **242** includes two links each. The two links are connected rotatably in horizontal directions to each other. The first arm **241** and the second arm **242** are connected to the upward/downward mover **250**. Each of the first arm **241** and the second arm **242** has one end connected rotatably in the horizontal directions to the upward/downward moving shaft **251** of the upward/downward mover **250**. The upward/downward mover **250** moves the first arm **241** and the second arm **242** upward/downward by moving the upward/downward moving shaft **251** in the vertical direction. The first arm **241** and the second arm **242**, and the upward/downward mover **250** are operated by driving forces of servomotors.

[0078] Each of the first arm **241** and the second arm **242** has another end on which the substrate-holding hand **230** is arranged. Specifically, the first blade support **231** is arranged on the another end of the first arm **241**. Also, the second blade support **232** is arranged on the another end of the second arm **242**. The first blade support **231** is connected rotatably in the horizontal directions to the first arm **241**. Also, the second blade support **232** is configured to be able to rotate in the horizontal directions with respect to the second arm **242**.

[0079] In the second embodiment, the first blade support **231** supports a proximal end of one blade **291**. Also, the second blade support **232** supports proximal ends of a plurality of blades **292**, **293**, **294** and **295**. Here, the blade **291** is an example of a first blade. The blades **292**, **293**, **294** and **295** are examples of a second blade.

[0080] The substrate-conveying robot **201** performs an operation of conveying one substrate **10** by using one blade **291** of the first blade support **231**, and an operation of simultaneously conveying a plurality of substrates **10** by using the plurality of blades **292**, **293**, **294** and **295** of the second blade support **232**.

[0081] In the second embodiment, the image capturer **260** is arranged in the second blade support **232**, which supports the plurality of blades **292**, **293**, **294** and **295**. Specifically, the image capturer **260** is arranged on the Z1-direction side, which is an upper side in the vertical direction, of the second blade support **232**. In other words, the image capturer **260** is arranged on the upper side with respect to all the plurality of blades **292**, **293**, **294** and **295** in the vertical direction.

[0082] The controller **70** detects the state of the inspection target based on the captured image P captured by the image capturer **260**. Processing for controlling detection of the state of the inspection target by using by the controller **70** is the same as the first embodiment.

[0083] Other configurations of the second embodiment are the same as the first embodiment above.

Advantages of Second Embodiment

[0084] In the second embodiment, the following advantages are obtained.

[0085] The substrate-holding hand **230** includes one blade **291** supported by the first blade support **231**, and the plurality of blades **292**, **293**, **294** and **295** supported by the second blade support **232**; and the image capturer **260** is arranged in the second blade support **232**, which supports the proximal ends of the plurality of blades **292**, **293**, **294** and **295**. Accordingly, in a case in which the plurality of substrates **10** are collectively conveyed by the plurality of blades **292**, **293**, **294** and **295** of the second blade support **232**, an image of the plurality of substrates **10** conveyed can be easily captured by the image capturer **260** arranged in the second blade support **232**. Consequently, in the case in which the plurality of substrates **10** are collectively conveyed, placement states of the plurality of substrates **10** conveyed can be easily detected based on the captured image P captured by the image capturer **260**. The other advantages of the second embodiment are similar to the aforementioned first embodiment.

Third Embodiment

[0086] The following description describes a configuration of a substrate processing system **300** according to a third embodiment of the present disclosure with reference to FIGS. **15** and **16**. Two image capturers **361** and **362** are provided in this third embodiment dissimilar to the aforementioned first embodiment in which one image capturer **60** is provided. The same components as those of the aforementioned first and second embodiments are denoted by the same reference numerals, and their description is omitted.

[0087] As shown in FIG. **15**, the substrate processing system **300** according to the third embodiment includes a substrate-conveying robot **301**. The substrate-conveying robot **301** includes the image capturers **361** and **362**. The substrate processing system **300** is an example of a semiconductor production apparatus system. The image capturers **361** and **362** are examples of first and second image capturers, respectively.

[0088] The image capturers **361** and **362** are two-dimensional cameras similar to the image capturer **60** in the first embodiment. Also, the image capturers **361** and **362** similarly capture images of the inspection target to detect the state of the inspection target.

[0089] As shown in FIG. **16**, in the third embodiment, the image capturer **361** is arranged in the first blade support **31**, and the image capturer **362** is arranged in the second blade support **32**. An arrangement of the image capturer **362** on the second blade support **32** is the same as the image capturer **60** in the first embodiment. In other words, in the third embodiment, in addition to the image capturer **362** arranged in the second blade support **32**, the additional image capturer **361** arranged in the first blade support **31** is provided to capture an image of the inspection target.

[0090] The image capturer **361** is arranged in the casing **31a** of the first blade support **31**. Specifically, while the image capturer **362** is arranged on one lateral side with respect to the first blade support **31**, the image capturer **361** is arranged on another lateral side, which is opposite to the one lateral side. Here, the one lateral side refers to the X1-direction side in FIG. **16**, and the another lateral side refers to the X2-direction side in FIG. **16**.

[0091] Also, the image capturer **362** is positioned at a position spaced away from the connection part **32c** similar to the image capturer **60** in the first embodiment. Also, the image capturer **361** is arranged in the first blade support **31** to be spaced at a spacing distance substantially equal to a spacing distance of the image capturer **362** away from the connection part **32c** in a direction opposite to the image capturer **362**. That is, the image capturers **361** and **362** are positioned at positions spaced at the substantially equal spacing distance away from the blades **91**, **92**, **93**, **94** and **95** in the X direction, which is a transverse direction of each of the plurality of blades **91**, **92**, **93**, **94** and **95**. Also, arrangements of the image capturers **361** and **362** in the Z-direction, which is the vertical direction, are substantially equal to each other. Accordingly, fields of view of the image capturers **361** and **362** are symmetrical to each other with respect to each of the plurality of blades **91**, **92**, **93**, **94** and **95**. Consequently, images of the plurality of blades **91**, **92**, **93**, **94** and **95** can be captured as a wide view captured image P from both sides in a leftward/rightward direction.

[0092] The controller **70** detects the state of the inspection target based on the captured images P captured by the image capturer **361** and the image capturer **362**. Processing for controlling detection of the state of the inspection target by using by the controller **70** is the same as the first embodiment.

[0093] Other configurations of the third embodiment are the same as the first embodiment above.

Advantages of Third Embodiment

[0094] In the third embodiment, the following advantages are obtained.

[0095] The substrate processing system **300** includes the image capturer **361** arranged in the first blade support **31**, and the image capturer **362** arranged in the second blade support **32** separately from the image capturer **361**. Accordingly, since the image capturer **361** is arranged in the first blade support **31**, and the image capturer **362** is arranged in the second blade support **32**, which operates independently of the first blade support **31**, the image capturer **361** and **362** can be

arranged to provide different fields of view from each other. Consequently, it is possible to capture a captured image P with a wider field of view as compared with a case of a single image capturer. As a result, it is possible to easily detect the state of the inspection target located over a wider area. The other advantages of the third embodiment are similar to the aforementioned first embodiment.

Modified Embodiments

[0096] Note that the embodiment disclosed this time must be considered as illustrative in all points and not restrictive. The scope of the present disclosure is not shown by the above description of the embodiments but by the scope of claims for patent, and all modifications (modified embodiments) within the meaning and scope equivalent to the scope of claims for patent are further included.

[0097] While the example in which the plurality of blades **91**, **92**, **93** and **94** are supported by the first blade support **31**, and one blade **95** is supported by the second blade support **32** has been shown in the aforementioned first and third embodiments; and one blade **291** is supported by the first blade support **231**, and the plurality of blades **95** are supported by the second blade support **232** has been shown in the aforementioned second embodiment, the present disclosure is not limited to these. In the present disclosure, the first blade support may support one first blade, and the second blade support may support one second blade. Alternatively, the first blade support may support a plurality of first blades, and the second blade support may support a plurality of second blades. The number of second blades supported by the second blade support on which the image capturer is arranged may be smaller than the number of first blades supported by the first blade support. Alternatively, the number of second blades may be greater than the number of first blades.

[0098] While the example in which the second blade support **32** supports the blade **95**, which is positioned at a middle position in the vertical direction in the plurality of blades **91**, **92**, **93**, **94** and **95** arranged adjacent to each other in the vertical direction has been shown in the aforementioned first and third embodiments, the present disclosure is not limited to this. In the present disclosure, the second blade support may support one blade that is positioned at the lowest position in the plurality of blades arranged adjacent to each other in the vertical direction. That is, only the one blade that is positioned at the lowest position in the plurality of blades may be operated separately from the other of the plurality of blades. Alternatively, the second blade support may support the blade that is positioned at the highest position in the vertical direction in the plurality of blades.

[0099] While the example in which the controller **70** for controlling operations of the substrate-conveying robot **101** executing detection of a state of the inspection target based on the captured image P has been shown in the aforementioned first embodiment, the present disclosure is not limited to this. In the present disclosure, the substrate-conveying robot may directly output the captured image that has been captured without executing detection of the state of the inspection target. In other words, a controller separate from the controller that controls the operations of the substrate-holding hand may execute detection of the state of the inspection target based on the captured image. For example, processing for controlling detection of the state of the inspection target may be executed in the higher-level control apparatus of the substrate processing system. Alternatively, a remote control system that is provided separately from the substrate processing system may execute detection of the state of the inspection target based on the captured image.

[0100] While the example in which the image capturer **60** is provided to move together with the second blade support **32** has been shown in the aforementioned first embodiment, the present disclosure is not limited to this. In the present disclosure, the image capturer may be arranged movably with respect to the second blade support. For example, the image capturer may be arranged to be able to change its image capture direction with respect to the second blade support.

[0101] While the example in which the image capturer **60** is arranged on the second blade support **32** with being spaced away from the connection part **32c** of the second blade support **32** has been shown in the aforementioned first embodiment, the present disclosure is not limited to this. In the present disclosure, the image capturer may be arranged near the connection part of the second blade support connected to the second blade.

[0102] While the example in which the placement states of the substrates **10** in the storage **20** are detected based on the captured image **P** of the substrates **10** captured with being stored in the storage **20** included in the processor **103** has been shown in the aforementioned first embodiment, the present disclosure is not limited to this. In the present disclosure, the placement states of the substrates in the storage may be detected by using an optical sensor arranged on the hand instead of the image capturer.

[0103] While the example in which the image capturer **60** is arranged on the lateral side with respect to the blades **91, 92, 93, 94** and **95** has been shown in the aforementioned first embodiment, and the example in which the image capturer **260** is arranged on the upper side with respect to the blades **292, 293, 294** and **295** has been shown in the aforementioned second embodiment, the present disclosure is not limited to these. In the present disclosure, the image capturer may be arranged on the lower side with respect to the second blade in the second blade support. Also, in a case in which the substrate-holding hand is linearly moved by the linear mover, the image capturer may be arranged on the upper or lower side with respect to the second blade. Also, in a case in which the substrate-holding hand is moved by the link mechanism including a robot arm, the image capturer may be arranged on the lateral or lower side with respect to the second blade support.

[0104] While the example in which an anomaly of the storage **20** as a component of the processor **103** is detected has been shown in the aforementioned first embodiment, the present disclosure is not limited to this. In the present disclosure, a state of a consumption degree of a consumable such as an electrode or solution used for processing the substrate in the processor as a component of the semiconductor production apparatus may be detected. Also, not only the processor for processing the substrate and the storage enclosure for storing the substrate but also components of a conveyor apparatus for conveying the substrates may be the inspection target. Also, foreign matters in the processor, the storage enclosure and the conveyor apparatus may be detected as the inspection target.

[0105] While the example in which the placement states of the substrates **10** placed in the storage **20** are detected has been shown in the aforementioned first embodiment, the present disclosure is not limited to this. In the present disclosure, the placement states of the substrates held in the substrate-holding hand may be detected based on the captured image. Also, an anomaly such as chipping or deformation of the substrate held in the substrate-holding hand may be detected. Also, a state of the substrate placed in the storage enclosure may be detected.

[0106] While the example in which two blade supports of the first blade support **31** and the second blade support **32** are provided has been shown in the aforementioned first embodiment, the present disclosure is not limited to this. In the present disclosure, three or more blade supports may be provided. In this configuration, the image capturer may be arranged only in the second blade support of the three or more blade supports, or in some of the three or more blade supports including at least the second blade support.

[0107] While the example in which one image capturer **60** is arranged in the second blade support **32** of the substrate-holding hand **30** has been shown in the aforementioned first embodiment, the present disclosure is not limited to this. In the present disclosure, a plurality of image capturers may be arranged in the second blade support.

[0108] The functions of the elements disclosed herein can be performed using circuits, including general-purpose processors, dedicated processors, integrated circuits, Application Specific Integrated Circuits (ASICs), conventional circuits, and/or combinations thereof, configured or programmed to perform the disclosed functions, circuitry or processing circuits, including general-purpose processors, dedicated processors, integrated circuits, ASICs (Application Specific Integrated Circuits), conventional circuits, and/or combinations thereof. A processor is considered a processing circuit or circuits because it contains transistors and other circuitry. In the present disclosure, a circuit, unit, or means is hardware that performs an enumerated function or is hardware programmed to perform an enumerated function. The hardware may be the hardware

disclosed herein or any other known hardware that is programmed or configured to perform the enumerated functions. If the hardware is a processor, which is considered a type of circuit, the circuit, means, or unit is a combination of hardware and software, and software is used to configure the hardware and/or processor.

Modes

[0109] It is understood by those skilled in the art that the exemplary embodiments described above are specific examples of the following aspects.

Mode Item 1

[0110] A semiconductor production apparatus system is a semiconductor production apparatus system for at least one of conveyance of substrates into and out of a semiconductor production apparatus for performing at least one of conveyance of the substrates, application of processing to the substrates and storing the substrates, the semiconductor production apparatus system including a substrate-holding hand including a first blade support for supporting a proximal end of a first blade on which one of the substrates is placed, and a second blade support for operating independently of the first blade support and for supporting a proximal end of a second blade on which another of the substrates is placed; an image capturer for capturing an image of an inspection target including at least one of the substrates, a component of the semiconductor production apparatus and the substrate-holding hand; and a controller for detecting a state of the inspection target based on the captured image captured by the image capturer, wherein the image capturer is arranged in the second blade support.

Mode Item 2

[0111] In the semiconductor production apparatus system according to mode item 1, the image capturer is moved together with the second blade support.

Mode Item 3

[0112] In the semiconductor production apparatus system according to mode item 1 or 2, the image capturer is arranged in the second blade support with being spaced away from a connection part between the second blade support and the second blade.

Mode Item 4

[0113] In the semiconductor production apparatus system according to mode item 3, the image capturer is arranged in a lateral part of the second blade support on an outside with respect to an edge of the second blade as viewed in a movement direction of the second blade in the conveyance of the substrates into and out of the semiconductor production apparatus with being spaced away from the connection part.

Mode Item 5

[0114] In the semiconductor production apparatus system according to any of mode items 1 to 4, the image capturer captures the image of the inspection target that includes at least the substrates with being stored in a storage included in the semiconductor production apparatus; and the controller detects, based on the captured image of the substrates captured with the substrates being stored in the storage, placement states of the substrates in the storage.

Mode Item 6

[0115] In the semiconductor production apparatus system according to any of mode items 1 to 5, the substrate-holding hand includes a plurality of first blades that are supported by the first blade support as the first blade, and one second blade, which is the second blade supported by the second blade support; and the image capturer is arranged in the second blade support, which supports the proximal end of the one second blade.

Mode Item 7

[0116] In the semiconductor production apparatus system according to any of mode items 1 to 6, a linear mover for linearly moving the substrate-holding hand is further provided; and the linear mover linearly moves the second blade support independently of the first blade support.

Mode Item 8

[0117] In the semiconductor production apparatus system according to mode item 7, an upward/downward mover for moving the linear mover upward/downward is further provided; and the upward/downward mover moves the substrate-holding hand upward/downward by moving the linear mover upward/downward.

Mode Item 9

[0118] In the semiconductor production apparatus system according to any of mode items 1 to 5, the substrate-holding hand includes one first blade, which is the first blade supported by the first blade support, and a plurality of second blades that are supported by the second blade support as the second blade; and the image capturer is arranged in the second blade support, which supports proximal ends of the plurality of second blades.

Mode Item 10

[0119] In the semiconductor production apparatus system according to any of mode items 1 to 9, the image capturer includes a first image capturer arranged in the first blade support, and a second image capturer arranged in the second blade support separately from the first image capturer.

Claims

1. A semiconductor production apparatus system for at least one of conveyance of substrates into and out of a semiconductor production apparatus for performing at least one of conveyance of the substrates, application of processing to the substrates and storing the substrates, the semiconductor production apparatus system comprising: a substrate-holding hand including a first blade support for supporting a proximal end of a first blade on which one of the substrates is placed, and a second blade support for operating independently of the first blade support and for supporting a proximal end of a second blade on which another of the substrates is placed; an image capturer for capturing an image of an inspection target including at least one of the substrates, a component of the semiconductor production apparatus and the substrate-holding hand; and a controller for detecting a state of the inspection target based on the captured image captured by the image capturer, wherein the image capturer is arranged in the second blade support.
2. The semiconductor production apparatus system according to claim 1, wherein the image capturer is moved together with the second blade support.
3. The semiconductor production apparatus system according to claim 1, wherein the image capturer is arranged in the second blade support with being spaced away from a connection part between the second blade support and the second blade.
4. The semiconductor production apparatus system according to claim 3, wherein the image capturer is arranged in a lateral part of the second blade support on an outside with respect to an edge of the second blade as viewed in a movement direction of the second blade in the conveyance of the substrates into and out of the semiconductor production apparatus with being spaced away from the connection part.
5. The semiconductor production apparatus system according to claim 1, wherein the image capturer captures the image of the inspection target that includes at least the substrates with being stored in a storage included in the semiconductor production apparatus; and the controller detects, based on the captured image of the substrates captured with the substrates being stored in the storage, placement states of the substrates in the storage.
6. The semiconductor production apparatus system according to claim 1, wherein the substrate-holding hand includes a plurality of first blades that are supported by the first blade support as the first blade, and one second blade, which is the second blade supported by the second blade support; and the image capturer is arranged in the second blade support, which supports the proximal end of the one second blade.
7. The semiconductor production apparatus system according to claim 1 further comprising a linear mover for linearly moving the substrate-holding hand, wherein the linear mover linearly moves the

second blade support independently of the first blade support.

8. The semiconductor production apparatus system according to claim 7 further comprising an upward/downward mover for moving the linear mover upward/downward, wherein the upward/downward mover moves the substrate-holding hand upward/downward by moving the linear mover upward/downward.

9. The semiconductor production apparatus system according to claim 1, wherein the substrate-holding hand includes one first blade, which is the first blade supported by the first blade support, and a plurality of second blades that are supported by the second blade support as the second blade; and the image capturer is arranged in the second blade support, which supports proximal ends of the plurality of second blades.

10. The semiconductor production apparatus system according to claim 1, wherein the image capturer includes a first image capturer arranged in the first blade support, and a second image capturer arranged in the second blade support separately from the first image capturer.
